

## Hardy Weinberg Equilibrium 2

### Hardy-Weinberg Equilibrium at an X-Linked Locus

X-linked loci are located on the X chromosome. Males have only one X chromosome, while females have two. This means the inheritance pattern and equilibrium dynamics are different from autosomal loci.

#### (I) HWE Is Not Reached After One Generation When Initial Allele Frequencies Differ Between Males and Females

For an autosomal locus, if allele frequencies differ between males and females at the start, one generation of random mating is enough to equalize them and establish HWE.

For an X-linked locus, this does NOT happen.

If the frequency of an allele (say  $X_A$ ) is different in males and females in the starting population, one generation of random mating will not restore equilibrium. Instead, the allele frequencies in males and females will oscillate and only gradually become equal over several generations. Equilibrium is reached only when allele frequencies become the same in both sexes and remain stable.

Why? Because males inherit their only X chromosome from their mother, while females inherit one X from each parent. This asymmetry causes the slower approach to equilibrium.

#### (II) Allele Frequency in Males Equals the Female Frequency from the Previous Generation

Let:

- $p_m$  = frequency of allele  $X_A$  in males
- $p_f$  = frequency of allele  $X_A$  in females

A male in the next generation receives his single X chromosome from his mother (fathers contribute a Y chromosome to sons). Therefore:

$$p_m (\text{next generation}) = p_f (\text{current generation})$$

In words: The allele frequency in males in the next generation is exactly the allele frequency in females in the current generation.

#### (III) Allele Frequency in Females Equals the Average of Male and Female Frequencies from the Previous Generation

A female in the next generation receives one X chromosome from her father and one from her mother.

- From her father: A father has only one X chromosome to pass to his daughters. So the frequency of XA in the father's contribution is p<sub>m</sub> (current generation).
- From her mother: The mother contributes an X allele drawn from the female population, with frequency p<sub>f</sub> (current generation).

Therefore, the frequency of XA in daughters is the average of the two:

$$p_f (\text{next generation}) = (p_m + p_f) / 2 \quad (\text{using current generation values})$$

In words: The allele frequency in females in the next generation is the average of the male and female frequencies in the current generation.

(IV) At HWE, Genotypic Frequencies Differ Between Males and Females Even When Allele Frequencies Are Equal

At equilibrium for an X-linked locus, the allele frequencies become equal in males and females. Let this common frequency be p for XA and q for Xa (where p + q = 1).

However, the genotypic frequencies are different between the sexes.

In males (who have only one X):

- Frequency of XA Y = p
- Frequency of Xa Y = q

In females (who have two X chromosomes):

- Frequency of XA XA = p<sup>2</sup>
- Frequency of XA Xa = 2pq
- Frequency of Xa Xa = q<sup>2</sup>

Example: If p = 0.6 and q = 0.4

| Genotype     | Frequency in Males | Frequency in Females |
|--------------|--------------------|----------------------|
| XA           | 0.6 (as XA Y)      | 0.36 (as XA XA)      |
| Xa           | 0.4 (as Xa Y)      | 0.16 (as Xa Xa)      |
| Heterozygote | not applicable     | 0.48 (as XA Xa)      |

Key Point: Males cannot be heterozygous for X-linked loci. Therefore, even with identical allele frequencies, the genotype frequencies remain different between sexes. This is a fundamental difference from autosomal loci, where genotype frequencies become identical in males and females at equilibrium.

## Summary

| Feature                                                   | Autosomal Locus           | X-Linked Locus                                 |
|-----------------------------------------------------------|---------------------------|------------------------------------------------|
| Time to equilibrium when starting p differs between sexes | 1 generation              | Several generations (with oscillations)        |
| Male allele frequency in next generation                  | Average of both sexes     | Equals current female frequency                |
| Female allele frequency in next generation                | Average of both sexes     | Average of current male and female frequencies |
| Genotype frequencies at equilibrium                       | Same in males and females | Different in males and females                 |

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## Modeling Selection as a force of evolution

### Selection in a One-Locus, Two-Allele Model

Natural selection occurs when individuals with different genotypes have different rates of survival, mating success, or reproduction. These differences are captured by the concept of fitness.

(I) Notation for Fitness:  $W_{11}$ ,  $W_{12}$ ,  $W_{22}$

For a gene with two alleles,  $A_1$  and  $A_2$ , there are three possible genotypes:

- $A_1A_1$
- $A_1A_2$
- $A_2A_2$

We assign a fitness value to each genotype. The standard notation is:

- $W_{11}$  = fitness of genotype  $A_1A_1$
- $W_{12}$  = fitness of genotype  $A_1A_2$
- $W_{22}$  = fitness of genotype  $A_2A_2$

Fitness reflects the contribution of a genotype to the next generation relative to other genotypes. It can be measured at different life stages, such as survival from zygote to adult, mating success, or fecundity.

### (II) Absolute Fitness

Absolute fitness is the actual number of offspring (or surviving individuals) produced by a genotype. It is measured in absolute terms, such as number of surviving adults per zygote.

### Numerical Example 1: Survival from Zygote to Adult

Suppose we start with 1000 zygotes of each genotype and measure how many survive to the adult stage:

| Genotype | Number of Zygotes | Number Surviving to Adult | Absolute Fitness (Survival Rate) |
|----------|-------------------|---------------------------|----------------------------------|
| A1A1     | 1000              | 1000                      | 1.00 (100% survival)             |
| A1A2     | 1000              | 900                       | 0.90 (90% survival)              |
| A2A2     | 1000              | 800                       | 0.80 (80% survival)              |

Here, absolute fitness is simply the proportion of zygotes that survive to become adults.

These absolute fitness values tell us the actual survival probability for each genotype.

### (III) Relative Fitness

Relative fitness compares the fitness of each genotype to the fitness of the most successful genotype. The most successful genotype is assigned a relative fitness of 1, and all others are expressed as a proportion of that maximum.

#### Numerical Example 1: Calculating Relative Fitness

From the absolute fitness values above:

- The highest absolute fitness is 1.00 (genotype A1A1).

We divide each absolute fitness by this maximum:

| Genotype | Absolute Fitness | Relative Fitness (w) |
|----------|------------------|----------------------|
| A1A1     | 1.00             | $1.00 / 1.00 = 1.00$ |
| A1A2     | 0.90             | $0.90 / 1.00 = 0.90$ |
| A2A2     | 0.80             | $0.80 / 1.00 = 0.80$ |

Relative fitness values are usually denoted as  $w_{11}$ ,  $w_{12}$ , and  $w_{22}$ .

#### Numerical Example 2: When the Heterozygote Is Most Fit

Now suppose the survival rates are different:

| Genotype | Absolute Fitness | Relative Fitness (w) |
|----------|------------------|----------------------|
| A1A1     | 0.85             | $0.85 / 0.95 = 0.89$ |
| A1A2     | 0.95             | $0.95 / 0.95 = 1.00$ |
| A2A2     | 0.70             | $0.70 / 0.95 = 0.74$ |

In this case, the heterozygote (A1A2) has the highest absolute fitness, so it is assigned a relative fitness of 1.

#### Numerical Example 3: Another Set of Survival Values

Now consider a scenario with the following survival rates:

| Genotype | Number of Zygotes | Number Surviving to Adult | Absolute Fitness (Survival Rate) |
|----------|-------------------|---------------------------|----------------------------------|
| A1A1     | 1000              | 600                       | 0.60 (60% survival)              |
| A1A2     | 1000              | 800                       | 0.80 (80% survival)              |
| A2A2     | 1000              | 900                       | 0.90 (90% survival)              |

To calculate relative fitness, we identify the highest absolute fitness, which is 0.90 (genotype A2A2). We then divide each absolute fitness by this maximum:

| Genotype | Absolute Fitness | Relative Fitness (w) |
|----------|------------------|----------------------|
| A1A1     | 0.60             | $0.60 / 0.90 = 0.67$ |
| A1A2     | 0.80             | $0.80 / 0.90 = 0.89$ |
| A2A2     | 0.90             | $0.90 / 0.90 = 1.00$ |

In this example, the A2A2 homozygote is the most fit, followed by the heterozygote, while the A1A1 homozygote has the lowest fitness.

#### (V) Why Use Relative Fitness?

Relative fitness is more useful than absolute fitness in population genetics models because:

1. Evolution depends on relative differences: Selection acts on the differences in fitness between genotypes, not on the absolute values. A genotype with absolute fitness 0.90 will increase in frequency if all others are below 0.90, but will decrease if another genotype has fitness 0.95.
2. Simplifies calculations: In the Hardy-Weinberg framework with selection, we typically use relative fitness to predict changes in allele frequencies from one generation to the next.
3. Removes scale: Absolute fitness can vary due to environmental conditions, but relative fitness captures the competitive advantage or disadvantage of each genotype.

#### (V) Summary Table of Examples

| Example   | Genotype | Absolute Fitness | Relative Fitness |
|-----------|----------|------------------|------------------|
| Example 1 | A1A1     | 1.00             | 1.00             |
|           | A1A2     | 0.90             | 0.90             |
|           | A2A2     | 0.80             | 0.80             |
| Example 2 | A1A1     | 0.85             | 0.89             |
|           | A1A2     | 0.95             | 1.00             |
|           | A2A2     | 0.70             | 0.74             |
| Example 3 | A1A1     | 0.60             | 0.67             |
|           | A1A2     | 0.80             | 0.89             |

#### (VI) Key Point

When modeling selection, we typically use relative fitness ( $w_{11}$ ,  $w_{12}$ ,  $w_{22}$ ) because the evolutionary outcome depends on how genotypes compare to one another, not on their

absolute performance. The most fit genotype is assigned a relative fitness of 1, and all others are scaled accordingly.

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### Modeling Dominance, Recessiveness, and Other Selection Scenarios

In selection models, we often express relative fitness in terms of selection coefficients to simplify the analysis. The selection coefficient measures the strength of selection against a genotype.

#### (I) The Selection Coefficient ( $s$ and $t$ )

The selection coefficient represents the proportional reduction in fitness of a genotype compared to the most fit genotype.

- $s$  is typically used for selection against a genotype (usually against a homozygote)
- $t$  is sometimes used when a second selection coefficient is needed (e.g., when both homozygotes are less fit than the heterozygote)

Range of values:

- $s$  ranges from 0 to 1
- $s = 0$  means no selection (fitness equal to the best genotype)
- $s = 1$  means complete selection (genotype contributes zero offspring; lethal)

#### (b) Table of Selection Scenarios

The table below shows how different relationships between alleles can be represented using fitness notation. In all cases, the most fit genotype(s) are assigned a relative fitness of 1.

| Scenario                    | Description                                                                                                                 | Fitness Notation                                     | Selection Coefficients                            | Example                                                  |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------|
| <b>Dominant Beneficial</b>  | The beneficial allele (A1) is dominant. A1A1 and A1A2 have equal highest fitness. The deleterious allele (A2) is recessive. | $W_{11} = 1$<br>$W_{12} = 1$<br>$W_{22} = 1 - s$     | $s$ = selection coefficient against A2A2          | If $s = 0.2$ :<br>A1A1 = 1.0<br>A1A2 = 1.0<br>A2A2 = 0.8 |
| <b>Recessive Beneficial</b> | The beneficial allele (A1) is recessive. Only A1A1 has highest fitness. The deleterious allele (A2) is dominant.            | $W_{11} = 1$<br>$W_{12} = 1 - s$<br>$W_{22} = 1 - s$ | $s$ = selection coefficient against A1A2 and A2A2 | If $s = 0.2$ :<br>A1A1 = 1.0<br>A1A2 = 0.8<br>A2A2 = 0.8 |

|                                                       |                                                                                                                       |                                                                     |                                                                                      |                                                                   |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Co-dominance<br/>(Additive)</b>                    | Neither allele is dominant. Fitness is intermediate for the heterozygote, exactly midway between the two homozygotes. | $W_{11} = 1$<br>$W_{12} = 1 - s/2$<br>$W_{22} = 1 - s$              | $s$ = selection coefficient against A2A2 (with A1A2 intermediate)                    | If $s = 0.2$ :<br>A1A1 = 1.0<br>A1A2 = 0.9<br>A2A2 = 0.8          |
| <b>Overdominance<br/>(Heterozygote Advantage)</b>     | The heterozygote has the highest fitness. Both homozygotes have reduced fitness.                                      | $W_{11} = 1 - s$<br>$W_{12} = 1$<br>$W_{22} = 1 - t$                | $s$ = selection coefficient against A1A1<br>$t$ = selection coefficient against A2A2 | If $s = 0.2, t = 0.4$ :<br>A1A1 = 0.8<br>A1A2 = 1.0<br>A2A2 = 0.6 |
| <b>Underdominance<br/>(Heterozygote Disadvantage)</b> | The heterozygote has the lowest fitness. Both homozygotes have higher fitness.                                        | $W_{11} = 1$<br>$W_{12} = 1 - s$<br>$W_{22} = 1 - t$<br><br>$s > t$ | $s$ = selection coefficient against A1A2                                             | If $s = 0.3$ :<br>A1A1 = 1.0<br>A1A2 = 0.7<br>A2A2 = 1.0          |

### (III) Explanation of Each Scenario

#### 1. Dominant Beneficial

The beneficial allele A1 is dominant. This means that having at least one copy of A1 confers the full fitness advantage. Only individuals who are A2A2 (recessive homozygote) suffer a fitness reduction. Examples include many resistance alleles where one copy provides protection.

#### 2. Recessive Beneficial

The beneficial allele A1 is recessive. Only individuals with two copies of A1 have the fitness advantage. All individuals with at least one copy of A2 have reduced fitness. This scenario is common for many deleterious recessive diseases, where the harmful allele A2 is dominant in its effect on fitness.

#### 3. Co-dominance (Additive)

Each copy of the beneficial allele A1 adds a fixed increment to fitness. The heterozygote has fitness exactly midway between the two homozygotes. This is also called additive selection because the effect of each allele is independent and additive.

#### 4. Overdominance (Heterozygote Advantage)

The heterozygote has higher fitness than either homozygote. This leads to a stable equilibrium where both alleles are maintained in the population (balancing selection). The classic example is sickle cell trait (A1A2) conferring resistance to malaria, while both homozygotes suffer disadvantages (anemia or malaria susceptibility) in areas with high incidence of malaria.

#### 5. Underdominance (Heterozygote Disadvantage)

The heterozygote has lower fitness than both homozygotes. This creates an unstable equilibrium. Populations tend to move toward fixation of one allele or the other, depending on initial allele frequencies. This scenario is less common in nature but can occur in cases of chromosomal rearrangements or hybrid incompatibility.

#### (d) Summary of Selection Coefficients

| Term                  | Symbol | Definition                                                                                 | Range  |
|-----------------------|--------|--------------------------------------------------------------------------------------------|--------|
| Selection coefficient | $s$    | Proportional reduction in fitness compared to the most fit genotype                        | 0 to 1 |
| Selection coefficient | $t$    | Second selection coefficient used when both homozygotes are less fit than the heterozygote | 0 to 1 |

Interpretation of values:

- $s = 0$ : No selection against the genotype
- $s = 0.1$ : 10% reduction in fitness
- $s = 0.5$ : 50% reduction in fitness
- $s = 1$ : Complete selection (genotype contributes no offspring; lethal)

#### (V) Key Point

The choice of fitness notation and selection coefficients depends on the biological relationship between the alleles. By using  $s$  and  $t$ , we can capture a wide range of selection scenarios with a compact mathematical representation, making it easier to predict changes in allele frequencies over generations.

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## Building a One-Locus, Two-Allele Model of Selection

We now build a model to predict how allele frequencies change from one generation to the next when selection is acting. We will relax the "no selection" assumption of Hardy-Weinberg Equilibrium and explicitly model selection as differences in survival from zygote to adult.

### (I) Assumptions of the Model

This model keeps most of the Hardy-Weinberg assumptions but relaxes the assumption of no selection. The assumptions are:

1. No mutation – Alleles do not change from one form to another.
2. No migration – No individuals enter or leave the population.
3. Large population size – Genetic drift is negligible.
4. Random mating – Individuals pair by chance with respect to the gene in question.
5. Selection acts at the zygote-to-adult stage – All genotypes start with equal numbers of zygotes, but they survive to adulthood at different rates. This is the only force causing evolution in this model.

### (II) Overview of the Life Cycle

We track the population through one generation using the following steps:

- Step 1: Gamete pool (allele frequencies  $p$  and  $q$ )
- Step 2: Random mating produces zygotes
- Step 3: Selection (survival differences from zygote to adult)
- Step 4: Adults (genotype frequencies after selection)
- Step 5: Adults produce gametes (next generation's gamete pool)
- Step 6: Calculate new allele frequency ( $p'$ ) and change ( $\Delta p$ )

We will now go through each step in detail.

### (III) Step 1: Gamete Pool (Current Generation)

In the current generation, the gamete pool (sperm and eggs) contains:

- Frequency of  $A_1 = p$
- Frequency of  $A_2 = q$
- Where  $p + q = 1$

These gametes come from the adults of the previous generation.

### (IV) Step 2: Random Mating Produces Zygotes

Because mating is random, zygotes are formed in Hardy-Weinberg proportions:

- A1A1 zygotes:  $p \times p = p^2$
- A1A2 zygotes:  $2 \times p \times q = 2pq$
- A2A2 zygotes:  $q \times q = q^2$

At this zygote stage, all genotypes are equally numerous according to these frequencies. There is no selection yet.

#### (V) Step 3: Selection Acts (Survival from Zygote to Adult)

Selection operates as differences in survival between the zygote and adult stages. Each genotype has a different probability of surviving to become a breeding adult.

We assign relative fitness values to represent survival rates:

- $w_{11}$  = relative fitness (survival rate) of genotype A1A1
- $w_{12}$  = relative fitness (survival rate) of genotype A1A2
- $w_{22}$  = relative fitness (survival rate) of genotype A2A2

These fitness values represent the proportion of zygotes that survive to adulthood. For example, if  $w_{11} = 0.9$ , then 90% of A1A1 zygotes survive to become adults.

After selection, the number (or frequency) of adults of each genotype is proportional to:

- A1A1 adults:  $p^2 \times w_{11}$
- A1A2 adults:  $2pq \times w_{12}$
- A2A2 adults:  $q^2 \times w_{22}$

#### (VI) Step 4: Calculate Adult Genotype Frequencies

The total number of adults after selection is proportional to the sum:

$$\text{Total} = (p^2 \times w_{11}) + (2pq \times w_{12}) + (q^2 \times w_{22})$$

This sum is called the mean fitness of the population, denoted as  $\bar{w}$  (pronounced "w bar").

$$\bar{w} = (p^2 \times w_{11}) + (2pq \times w_{12}) + (q^2 \times w_{22})$$

We then divide each genotype's contribution by  $\bar{w}$  to get the actual frequencies among adults:

- Frequency of A1A1 among adults =  $(p^2 \times w_{11}) / \bar{w}$
- Frequency of A1A2 among adults =  $(2pq \times w_{12}) / \bar{w}$
- Frequency of A2A2 among adults =  $(q^2 \times w_{22}) / \bar{w}$

These adult frequencies add up to 1.

### (VII) Step 5: Adults Produce Gametes (Next Generation's Gamete Pool)

The adults now produce gametes to form the next generation. The frequency of A1 in the gametes they produce is simply the frequency of A1 among the adults.

To calculate this:

- All alleles from A1A1 adults are A1
- Half the alleles from A1A2 adults are A1
- A2A2 adults contribute no A1 alleles

Therefore:

$$p' = (1 \times \text{frequency of A1A1 among adults}) + (0.5 \times \text{frequency of A1A2 among adults})$$

$$p' = (p^2 \times w_{11}) / \bar{w} + (0.5 \times 2pq \times w_{12}) / \bar{w}$$

Simplify  $0.5 \times 2pq = pq$ :

$$p' = (p^2 \times w_{11}) / \bar{w} + (pq \times w_{12}) / \bar{w}$$

Factor out p from the numerator:

$$p' = p \times (p \times w_{11} + q \times w_{12}) / \bar{w}$$

This  $p'$  is the frequency of allele A1 in the next generation's gamete pool.

### (VIII) Step 6: Calculate the Change in Allele Frequency ( $\Delta p$ )

The change in allele frequency from one generation to the next is:

$$\Delta p = p' - p$$

Substitute  $p'$ :

$$\Delta p = [p \times (p \times w_{11} + q \times w_{12}) / \bar{w}] - p$$

Factor out p:

$$\Delta p = p \times [(p \times w_{11} + q \times w_{12}) / \bar{w} - 1]$$

Write 1 as  $\bar{w} / \bar{w}$ :

$$\Delta p = p \times [(p \times w_{11} + q \times w_{12} - \bar{w}) / \bar{w}]$$

**This is the equation for  $\Delta p$ :**

$$\Delta p = (p / \bar{w}) \times (p \times w_{11} + q \times w_{12} - \bar{w})$$

(IX) Numerical Example: Putting It All Together

Let's work through a concrete example.

Given:

- Starting allele frequency:  $p = 0.6$ ,  $q = 0.4$
- Fitness values (survival rates):
  - $w_{11} = 1.0$  (A1A1: 100% survive)
  - $w_{12} = 0.9$  (A1A2: 90% survive)
  - $w_{22} = 0.8$  (A2A2: 80% survive)

Step 1: Zygote frequencies

- A1A1:  $p^2 = 0.36$
- A1A2:  $2pq = 0.48$
- A2A2:  $q^2 = 0.16$

Step 2: Mean fitness

$$\begin{aligned}\bar{w} &= (0.36 \times 1.0) + (0.48 \times 0.9) + (0.16 \times 0.8) \\ \bar{w} &= 0.36 + 0.432 + 0.128 = 0.92\end{aligned}$$

Step 3: New allele frequency

$$\begin{aligned}p' &= p \times (p \times w_{11} + q \times w_{12}) / \bar{w} \\ p' &= 0.6 \times (0.6 \times 1.0 + 0.4 \times 0.9) / 0.92 \\ p' &= 0.6 \times (0.6 + 0.36) / 0.92 \\ p' &= 0.6 \times 0.96 / 0.92 \\ p' &= 0.576 / 0.92 = 0.626\end{aligned}$$

Step 4: Change in allele frequency

$$\Delta p = p' - p = 0.626 - 0.6 = 0.026$$

The frequency of A1 increased by 0.026 (from 0.6 to 0.626) in one generation because it has a fitness advantage.

(X) Summary of Key Equations

| Stage                            | Equation                                                                      |
|----------------------------------|-------------------------------------------------------------------------------|
| Mean fitness                     | $\bar{w} = (p^2 \times w_{11}) + (2pq \times w_{12}) + (q^2 \times w_{22})$   |
| Next generation allele frequency | $p' = p \times (p \times w_{11} + q \times w_{12}) / \bar{w}$                 |
| Change in allele frequency       | $\Delta p = p \times (p \times w_{11} + q \times w_{12} - \bar{w}) / \bar{w}$ |

## (XI) Key Point

This model shows how selection changes allele frequencies across generations. The direction and speed of evolution depend on:

- Current allele frequencies ( $p$  and  $q$ )
- The fitness of each genotype ( $w_{11}$ ,  $w_{12}$ ,  $w_{22}$ )
- The mean fitness of the population ( $\bar{w}$ )

The equation  $\Delta p$  tells us whether allele  $A_1$  is increasing ( $\Delta p > 0$ ), decreasing ( $\Delta p < 0$ ), or stable ( $\Delta p = 0$ ).

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### Behavior of the Selection Model – What Affects the Rate of Change?

Now that we have built the selection model, we can explore how different factors influence the rate at which allele frequencies change. We will also examine how dominance relationships shape the trajectory of a new beneficial mutation.

#### (I) Factors Affecting the Rate of Change ( $\Delta p$ )

Two main factors determine how quickly allele frequencies change under selection:

##### (1) The Magnitude of Fitness Differences

The larger the fitness differences between genotypes, the faster the change in allele frequency.

- If selection is strong (e.g.,  $w_{11} = 1.0$ ,  $w_{12} = 1.0$ ,  $w_{22} = 0.5$ ),  $\Delta p$  will be large each generation.
- If selection is weak (e.g.,  $w_{11} = 1.0$ ,  $w_{12} = 0.99$ ,  $w_{22} = 0.98$ ),  $\Delta p$  will be small each generation.

In our  $\Delta p$  equation, this is captured by the differences ( $w_{11} - w_{12}$ ) and ( $w_{12} - w_{22}$ ). Larger differences mean larger  $\Delta p$ .

##### (2) Dominance Relationships

The way alleles interact (dominance, recessiveness, etc.) affects the trajectory of allele frequency change. The same fitness difference can lead to very different patterns depending on whether the beneficial allele is dominant or recessive.

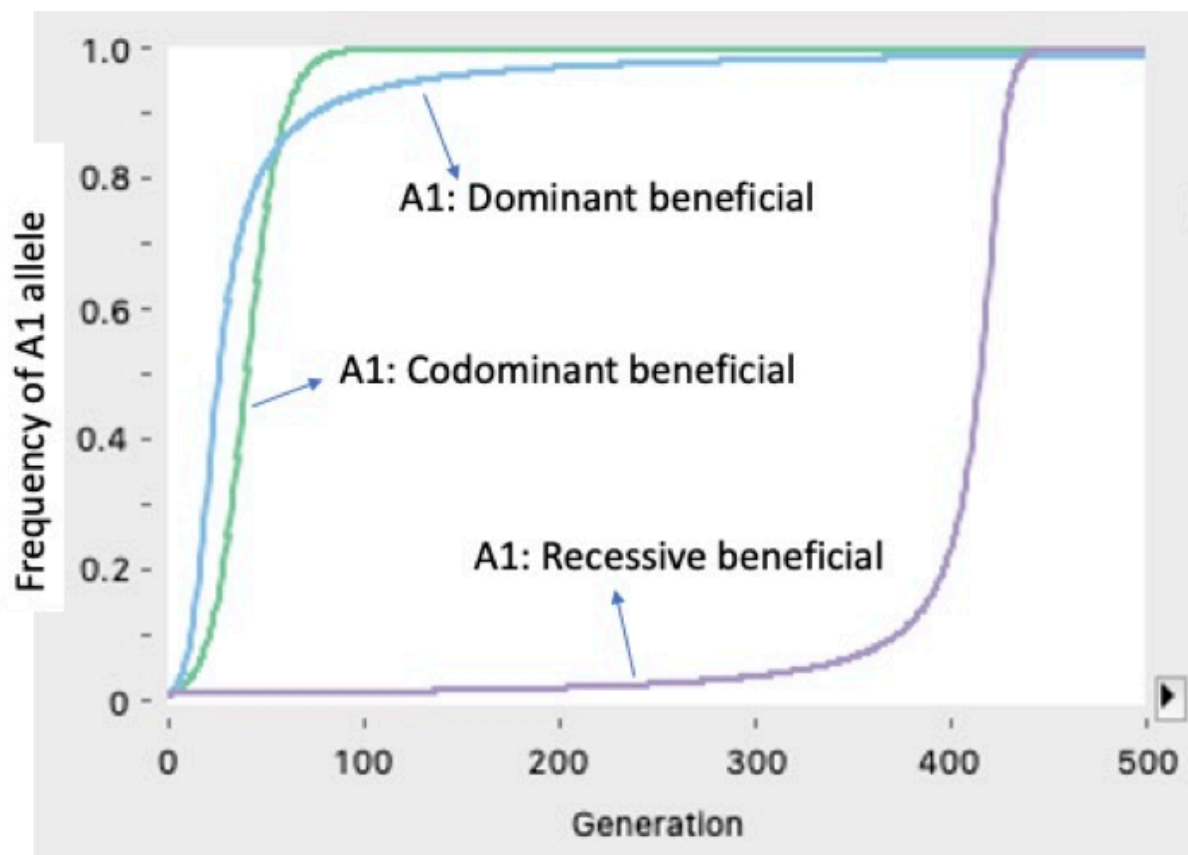
We will explore these ideas using a thought experiment.

## (II) Thought Experiment: The Fate of a New Beneficial Mutation

Imagine a population that initially contains only the A2 allele. The population is fixed for A2, so  $p = 0$  and  $q = 1$ .

Suddenly, a new mutation occurs in one individual, creating a single copy of a new allele A1. This new allele is beneficial – it increases survival. However, it starts at a very low frequency ( $p$  is close to 0).

What happens next depends on how the beneficial allele behaves with respect to dominance.



## (III) Scenario 1: Dominant Beneficial (Recessive Deleterious)

In this case:

- $w_{11} = 1$  (A1A1 has highest fitness)
- $w_{12} = 1$  (A1A2 has same high fitness – A1 is dominant)
- $w_{22} = 1 - s$  (A2A2 has lower fitness)

What happens to the new A1 allele?

Because A1 is dominant, even individuals with just one copy (heterozygotes) have the full fitness advantage. When the mutation first appears, it exists only in heterozygotes (since it is

rare, it is unlikely to be in a homozygote). These heterozygotes survive and reproduce better than the A2A2 individuals that make up most of the population.

Trajectory:

- The frequency of A1 increases rapidly at first because heterozygotes have a strong advantage.
- As A1 becomes more common, many individuals become A1A1 homozygotes, but this does not change the fitness advantage (since both A1A1 and A1A2 have fitness = 1).
- However, as the frequency of A2 declines, the opportunity for selection diminishes. The last steps toward fixation ( $p = 1$ ) are very slow.

Why the slow finish?

When A1 is very common and A2 is very rare, almost all remaining A2 alleles exist in heterozygotes (A1A2). These heterozygotes have the same high fitness as A1A1 individuals. Therefore, there is almost no selection against the A2 allele when it is rare. It takes a very long time (mathematically infinite) to completely eliminate the last copies of A2.

Key point: In dominant beneficial selection, the beneficial allele rises quickly from low frequency but takes a long time to reach fixation.

#### (IV) Scenario 2: Recessive Beneficial (Dominant Deleterious)

In this case:

- $w_{11} = 1$  (A1A1 has highest fitness – A1 is recessive)
- $w_{12} = 1 - s$  (A1A2 has reduced fitness – the deleterious A2 is dominant)
- $w_{22} = 1 - s$  (A2A2 also has reduced fitness)

What happens to the new A1 allele?

The new A1 allele is beneficial only when it is in a homozygote (A1A1). But when the mutation first appears, it exists only in heterozygotes (A1A2). These heterozygotes do NOT have the fitness advantage. In fact, they have the same reduced fitness as A2A2 individuals (since the deleterious A2 allele is dominant).

Trajectory:

- Initially, the A1 allele is "hidden" in heterozygotes and is subject to selection against it (because heterozygotes have lower fitness).
- The frequency of A1 may actually decrease at first! This is called the "lag phase."
- Only when the allele becomes common enough that A1A1 homozygotes start to appear does the beneficial effect become visible.
- Once A1A1 homozygotes appear, they have a high fitness advantage, and A1 increases rapidly.
- After this point, the approach to fixation is faster than in the dominant beneficial case because homozygotes are the ones that carry the advantage.

Key point: In recessive beneficial selection, the beneficial allele initially struggles (or even declines) because it is hidden in heterozygotes. After a delay, it increases rapidly once homozygotes appear.

#### (V) Scenario 3: Co-dominance (Additive)

In this case:

- $w_{11} = 1$  (A1A1 has highest fitness)
- $w_{12} = 1 - s/2$  (A1A2 has intermediate fitness)
- $w_{22} = 1 - s$  (A2A2 has lowest fitness)

What happens to the new A1 allele?

Each copy of A1 adds a fixed increment to fitness. Heterozygotes have fitness exactly midway between the two homozygotes.

Trajectory:

- The new A1 allele, when rare, exists in heterozygotes. These heterozygotes have a small but real fitness advantage over A2A2 individuals.
- Therefore, A1 increases steadily from the beginning.
- The increase is neither as fast initially as the dominant beneficial case, nor as delayed as the recessive beneficial case.
- The approach to fixation is smooth and steady.

Key point: In co-dominance (additive selection), the beneficial allele increases at a constant rate (in terms of logit or odds) and there is no lag phase or slow finish.

Here is the revised section for overdominance and underdominance, presented without the thought experiment and with emphasis on equilibrium properties and fitness landscapes.

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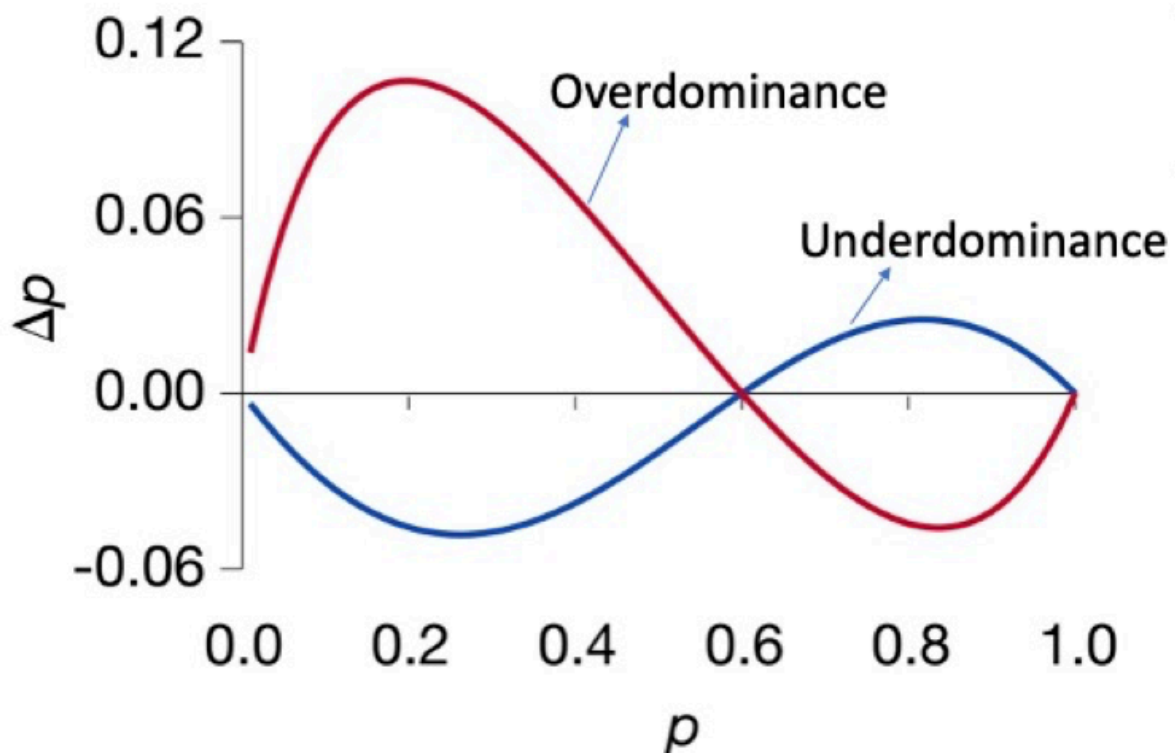
#### Overdominance and Underdominance – Equilibrium Properties

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Unlike the previous scenarios where a beneficial allele eventually goes to fixation (or is lost), overdominance and underdominance create internal equilibrium points where both alleles are maintained. However, these two scenarios behave very differently in terms of stability and the effect of initial allele frequency.



## $\Delta p$ as a function of $p$



### (VI) Overdominance (Heterozygote Advantage)

In overdominance, the heterozygote has the highest fitness. Both homozygotes have lower fitness.

Fitness notation:

- $w_{11} = 1 - s$  ( $A_1A_1$  has reduced fitness)
- $w_{12} = 1$  ( $A_1A_2$  has highest fitness)
- $w_{22} = 1 - t$  ( $A_2A_2$  has reduced fitness)

Here,  $s$  and  $t$  are positive selection coefficients (between 0 and 1) representing the fitness disadvantage of each homozygote.

#### (i) Globally Stable Equilibrium

Overdominance creates an internal equilibrium point where  $\Delta p = 0$ . This equilibrium frequency is given by:

$$\hat{p} = t / (s + t)$$

At this frequency, the advantage of being a heterozygote is exactly balanced by the disadvantage of producing homozygotes.

Key property: This equilibrium is globally stable. This means:

- No matter what the initial allele frequency is (as long as both alleles are present), the population will always move toward this equilibrium frequency.
- If the starting frequency is below  $\hat{p}$ , A1 increases.
- If the starting frequency is above  $\hat{p}$ , A1 decreases.
- Once at equilibrium, the population remains there indefinitely (unless other evolutionary forces act).

The population will converge to this same equilibrium whether it starts with A1 at 0.01, 0.5, or 0.99.

#### (ii) Relationship to Mean Fitness

The equilibrium point in overdominance corresponds to the maximum mean fitness ( $\bar{w}$ ) of the population.

- If you calculate  $\bar{w} = p^2 w_{11} + 2pq w_{12} + q^2 w_{22}$  across all possible  $p$  values, the highest value occurs exactly at  $\hat{p} = t/(s+t)$ .
- Natural selection acts to maximize the mean fitness of the population. Therefore, the population is "pulled" toward this point.
- This is why overdominance maintains genetic diversity – the population is most fit when both alleles are present.

Example:

If  $s = 0.2$  and  $t = 0.4$ , then:

$$\hat{p} = 0.4 / (0.2 + 0.4) = 0.4 / 0.6 = 0.67$$

The population will always move toward  $p = 0.67$ , regardless of where it started.

#### (VII) Underdominance (Heterozygote Disadvantage)

In underdominance, the heterozygote has the lowest fitness. Both homozygotes have higher fitness.

Fitness notation:

- $w_{11} = 1$  (A1A1 has high fitness)
- $w_{12} = 1 - s$  (A1A2 has reduced fitness – heterozygote disadvantage)
- $w_{22} = 1$  (A2A2 has high fitness)

Here,  $s$  is a positive selection coefficient (between 0 and 1) representing the fitness disadvantage of the heterozygote.

#### (i) Unstable Equilibrium

Underdominance also creates an internal equilibrium point where  $\Delta p = 0$ .

This equilibrium is unstable. This means:

- What happens to the allele frequency depends entirely on the starting point.
- If the starting frequency is above  $\hat{p}$ , the population will move toward fixation of A1 ( $p = 1$ ).
- If the starting frequency is below  $\hat{p}$ , the population will move toward loss of A1 ( $p = 0$ ).
- If the population starts exactly at  $\hat{p}$ , it is at an unstable point. Any small random fluctuation (due to genetic drift) will push it away, and it will then move toward one of the homozygote fixations.

## (ii) Relationship to Mean Fitness

The internal equilibrium point in underdominance corresponds to the minimum mean fitness ( $\bar{w}$ ) of the population.

- If you calculate  $\bar{w} = p^2 w_{11} + 2pq w_{12} + q^2 w_{22}$  across all possible  $p$  values, the lowest value occurs exactly at  $\hat{p}$  (0.5 in the symmetrical case).
- The two homozygote fixations ( $p = 0$  and  $p = 1$ ) represent local fitness peaks, but neither is necessarily the global maximum if the fitness values are equal.

Key point: Underdominance creates a "valley" in the fitness landscape. Populations that find themselves on one side of the valley will go to the nearest peak. The population may end up at a fitness peak that is not globally highest if the peaks are of different heights.

## (iii) Dependence on Starting Point

Unlike overdominance, underdominance has no single predictable outcome. The final state depends on initial conditions:

This means two populations with the same fitness values but different starting frequencies can end up in different states. One population may fix A1 while another loses it entirely.

| Feature                      | Overdominance                      | Underdominance                                          |
|------------------------------|------------------------------------|---------------------------------------------------------|
| Heterozygote fitness         | Highest                            | Lowest                                                  |
| Homozygote fitness           | Lower than heterozygote            | Higher than heterozygote                                |
| Internal equilibrium         | Present                            | Present                                                 |
| Stability                    | Globally stable                    | Unstable                                                |
| Effect of starting point     | Always reaches same equilibrium    | Outcome depends on starting point                       |
| Mean fitness at equilibrium  | Maximum                            | Minimum                                                 |
| Maintains genetic variation? | Yes (balanced polymorphism)        | No (population moves to fixation of one allele)         |
| Fate of new mutation         | Increases to equilibrium frequency | Usually lost unless it reaches high frequency by chance |

### (VIII) Key Point

Overdominance and underdominance both create internal equilibrium points where selection alone does not change allele frequency, but they behave in opposite ways. In overdominance, the equilibrium is globally stable and represents the peak of population fitness – the population always converges to it. In underdominance, the equilibrium is unstable and represents a fitness valley – the final outcome depends on the starting frequency, and populations can get "stuck" at a local fitness peak that is not the highest possible.